Optimization

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Contents

Section 1

Definitions

Constrained choice

Optimization selects the *best feasible point*.

- The objective assigns a value to every feasible point.
- Constraints determine the feasible set.

Example: maximize a utility function over budget-feasible set of product bundles.

Optimization problems

General form. Given a feasible set $S \subseteq \mathbb{R}^n$ and an objective $f : S \to \mathbb{R}$,

$$\max_{x \in S} f(x).$$

Maximizer and maximum. $x^* \in S$ is a **maximizer** if $f(x) \leq f(x^*)$ for every $x \in S$. The set of all maximizers is $\arg\max_{x \in S} f(x)$; when this set is nonempty, the **maximum value** is $\max_{x \in S} f(x) = f(x^*)$.

The optimizer is a point (or a set of points); the optimized value is a number. Several choices can maximize the same objective, but the maximum value is unique.

Video: Steve Brunton, *The anatomy of an optimization problem*.

Four questions in an optimization problem

Once the objective and feasible set are fixed, separate four logically different questions.

Existence

Is $\arg\max_{x\in S} f(x)$ nonempty?

Globality and uniqueness

When is a local optimum global, and when is the optimizer unique?

Optimality conditions

Which conditions are necessary, and when are they sufficient?

Sensitivity

How do the optimal choice and optimized value change with parameters?

> Compactness and continuity address existence; FOC, SOC, Lagrange and KKT give optimality conditions; curvature connects local and global conclusions; comparative statics and the envelope theorem study parameter changes.

Consumer and firm problems

The same template separates three objects: the **choice variable**, the **feasible set**, and the **objective**.

Consumer choice

$$\max_{x \in \mathbb{R}^n_+} u(x) \quad \text{s.t.} \quad p \cdot x \leq w.$$

- x: consumption bundle.
- $p \cdot x \leq w$: budget feasibility.
- $u(x)$: objective to maximize.

Firm choice

$$\max_{q \geq 0} \; pq - c(q).$$

- q: output choice.
- $q \geq 0$: feasible production scale.
- $pq - c(q)$: profit objective.

When u represents the consumer's preferences, the preferences are convex iff u is quasiconcave; strict quasiconcavity makes an attained maximizer unique on the convex budget set.

Further reading: Border, *Preference and Demand Examples*; Border, *Examples of Cost and Production Functions*.

Data fitting and allocation

The same notation covers estimation and allocation; minimizing C is equivalent to maximizing $-C$.

Least squares fit

$$\min_{\beta \in \mathbb{R}^k} \|y - X\beta\|^2$$

- β: model coefficients.
- $\mathbb{R}^k$: unrestricted feasible set.
- $\|y - X\beta\|^2$: prediction error.

Resource allocation

$$\min_{a \in \mathbb{R}^m_+} C(a)$$

$$\text{s.t.} \sum_{i=1}^{m} a_i = R.$$

- a_i: amount assigned to use i.
- $\sum_i a_i = R$: resource feasibility.
- $C(a)$: cost to minimize.

Local and global extrema

Let $x^* \in S$.

Global maximum $f(x) \leq f(x^*)$ for every $x \in S$.

Local maximum $f(x) \leq f(x^*)$ for every feasible x near x^*.

Global minimum $f(x) \geq f(x^*)$ for every $x \in S$.

Local minimum $f(x) \geq f(x^*)$ for every feasible x near x^*.

$$x^* \text{ global max} \Rightarrow x^* \text{ local max}, \qquad x^* \text{ global min} \Rightarrow x^* \text{ local min}.$$

Hence a sufficient condition for a global extremum is sufficient for the corresponding local extremum, while a necessary condition for a local extremum is necessary for the corresponding global extremum. The converses need not hold.

Video: Khan Academy, *Multivariable maxima and minima*.

Local vs. global extrema

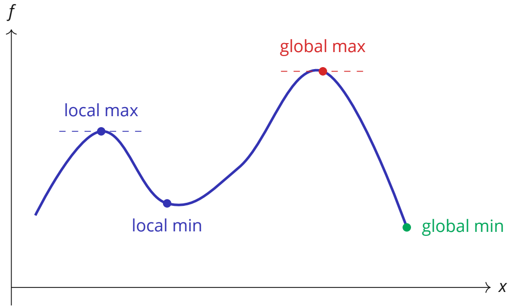

Figure 1 · Local and global extrema.

Local extrema compare nearby feasible points; a global extremum may occur at a boundary point.

Local optima are global

Convexity of the feasible set keeps every segment from a local candidate to another feasible point inside the feasible set.

Proposition. Let $S \subseteq \mathbb{R}^n$ be convex and let $f : S \to \mathbb{R}$.

(i) If f is convex, every local minimizer is global. If f is strictly convex, f has at most one minimizer.

(ii) If f is concave, every local maximizer is global. If f is strictly concave, f has at most one maximizer.

Further reading: MIT 6.253, *Convex Analysis and Optimization*, p. 59; Boyd and Vandenberghe, *Convex Optimization*, §4.2.

A constrained maximum

Maximize $f(x)$ over $x \in [-1, 2]$.

- The unconstrained maximizer can lie outside the feasible set.
- The constrained maximizer can then sit on the boundary.

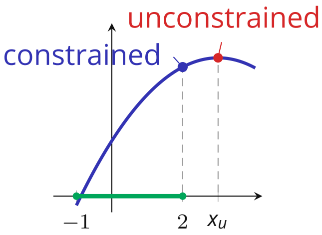

Figure 2 · The feasible interval is green.

The constrained maximum occurs at the boundary point $x = 2$.

Useful transformations

- Minimizers of f are maximizers of $-f$.
- If g is strictly increasing on the range of f, then

$$\arg\max_{x\in S} f(x) = \arg\max_{x\in S} g(f(x)).$$

Log trick. For $f(x) = A\prod_i x_i^{\alpha_i} > 0$, $\ln f = \ln A + \sum_i \alpha_i \ln x_i$. Maximizing a positive likelihood $L(\theta)$ is likewise equivalent to maximizing $\ell(\theta) = \ln L(\theta)$.

A strictly increasing transformation preserves the optimizer set but changes the numerical value of the objective. This distinction matters later when we separate comparative statics of choices from derivatives of the value function.

Further reading: Boyd, Ch. 20, §§20.3.3–20.3.5.

Section 2

Existence, FOC & SOC

Does an optimization problem attain its optimum?

Here, existence means that at least one feasible choice attains the largest objective value:

$$\arg\max_{x\in S} f(x) \neq \varnothing.$$

- Not every optimization problem has a maximizer: $\max_{x\in\mathbb{R}} x$ has none.
- If $S \subseteq \mathbb{R}^n$ is nonempty and compact and $f : S \to \mathbb{R}$ is continuous, the Extreme Value Theorem (Weierstrass) guarantees a maximizer $x^* \in S$.
- Concavity serves a different role: it helps turn local conditions into global conclusions.

Logical order: first ask whether a solution exists; then derive optimality conditions and distinguish necessity from sufficiency; then use curvature, or compare the candidate values directly, to establish global optimality and uniqueness.

Further reading: Boyd, §29.42+ and §30.2; Carter, Ch. 6, pp. 601–602; §6.2, pp. 609–621.

Video: Mike, the Mathematician, *Continuous Functions on Compact Sets and the Extreme Value Theorem*.

Stationarity

FONC. If f is **differentiable** at x^* and x^* is an **interior** local extremum, then

$$\nabla f(x^*) = 0.$$

1. Solve $\nabla f(x) = 0$ to find interior critical points.
2. Check the **boundary** and **non-differentiable** points separately.
3. Compare objective values when several feasible possibilities remain.

This is the first-order local-linearization result from Differentiation: at an unconstrained interior extremum, every direction is feasible, so every directional derivative must be zero. Stationarity is necessary; existence and optimality require separate checks.

Video: Krista King, *Critical points of multivariable functions*.

At an interior critical point

Write a nearby point as $x^* + h$. The second-order Taylor expansion is

$$f(x^* + h) - f(x^*) = \nabla f(x^*)^\top h + \frac{1}{2} h^\top H_f(x^*) h + o(\|h\|^2).$$

At an interior critical point, $\nabla f(x^*) = 0$, so

$$f(x^* + h) - f(x^*) = \frac{1}{2} Q(h) + o(\|h\|^2), \qquad Q(h) := h^\top H_f(x^*) h.$$

Local maximum. Necessary: $H(x^*) \preceq 0$.
Sufficient: $H(x^*) \prec 0$.

Local minimum. Necessary: $H(x^*) \succeq 0$.
Sufficient: $H(x^*) \succ 0$.

An indefinite Hessian gives a saddle point. A semidefinite Hessian can leave the second-order test inconclusive.

Video: Dr. Trefor Bazett, *Multi-variable optimization and the second derivative test*.

Local test versus global test

Suppose $S \subseteq \mathbb{R}^n$ is convex and f is twice continuously differentiable on an open set containing S. The same Hessian appears in both tests; the quantifier is what changes.

Convexity on S

$H_f(x) \succeq 0$ **for every** $x \in S \Rightarrow f$ is convex on S.

If $H_f(x) \succ 0$ **for every** $x \in S$, then f is strictly convex on S.

Concavity on S

$H_f(x) \preceq 0$ **for every** $x \in S \Rightarrow f$ is concave on S.

If $H_f(x) \prec 0$ **for every** $x \in S$, then f is strictly concave on S.

Apply the earlier proposition. Once the Hessian signs establish convexity or concavity on all of S, the local-to-global and uniqueness conclusions from "Local optima are global" apply.
Definiteness of $H_f(x^*)$ at the single point x^* gives only a local conclusion, and a boundary optimum need not satisfy $\nabla f(x^*) = 0$.

Further reading: *Video: Steve Brunton, Convexity 101*. Boyd, Ch. 21.

A quadratic objective

On $(x_1, x_2) \in [0,4]^2$, let

$$f(x_1, x_2) = 2x_1x_2 - x_1^2 - 2x_2^2 + 3x_1 - 2x_2.$$

First-order conditions

$$\nabla f(x) = 0 \quad \Rightarrow \quad (x_1^*, x_2^*) = (2, \tfrac{1}{2}).$$

Second-order test

$$H_f = \begin{pmatrix} -2 & 2 \\ 2 & -4 \end{pmatrix}.$$

$D_1 = -2 < 0$ and $D_2 = 4 > 0$, so $H_f \prec 0$.

> The constant Hessian is negative definite, so f is strictly concave on $[0,4]^2$. Hence $(2, \frac{1}{2})$ is the unique global maximizer. Since f is quadratic, its second-order Taylor formula has no higher-order remainder.

Section 3

Equality constraints

The Lagrangian

Problem. $\max_{x \in \mathbb{R}^n} f(x)$ subject to $g_j(x) = b_j$, $j = 1, \ldots, m$.

$$\mathcal{L}(x, \lambda) = f(x) + \sum_{j=1}^{m} \lambda_j \big(b_j - g_j(x)\big).$$

- If the equality-constraint gradients are linearly independent at a constrained local optimum, then $\nabla_x \mathcal{L} = 0$, together with $g_j(x) = b_j$.
- Substitute the constraint directly when this is simpler.
- If f is concave and the equality constraints are affine, the FOCs are sufficient for a global maximum.

Further reading: Boyd, Ch. 18.

Video: Dr. Trefor Bazett, *Lagrange Multipliers: Geometric Meaning and Example*.

Tangency of level curve and constraint

At a regular interior constrained optimum with $\nabla f(x^*) \neq 0$, the objective level curve and the constraint curve are tangent:

$$\nabla f(x^*) = \lambda^* \nabla g(x^*).$$

- The gradients are collinear normals; their directions may agree or oppose.
- Where $\nabla g(x^*) \neq 0$, this condition pins down the multiplier λ^*; its interpretation comes later in this section.

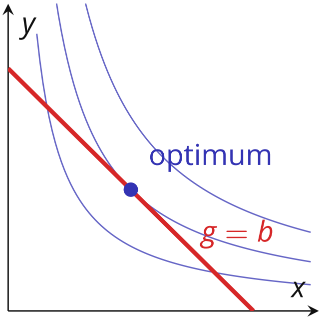

Figure 3 · An objective level curve and a constraint curve.

At the regular optimum shown, tangency is a first-order necessary condition.

Second-order conditions along feasible directions

Feasible directions and restricted curvature

Consider

$$\max_x f(x) \qquad \text{s.t.} \qquad h(x) = 0,$$

where $h : \mathbb{R}^n \to \mathbb{R}^m$. Let (x^*, λ^*) satisfy the regular first-order conditions and write

$$H_{\mathcal{L}} = \nabla^2_{xx}\mathcal{L}(x^*, \lambda^*).$$

A first-order feasible direction d is tangent to the constraint set when

$$Dh(x^*)d = 0.$$

Second-order conditions along feasible directions

Curvature on the tangent space

Necessary for max $d^\top H_{\mathcal{L}} d \leq 0$ for every tangent direction d.

Sufficient for max $d^\top H_{\mathcal{L}} d < 0$ for every nonzero tangent direction d.

Unconstrained SOC test curvature in every direction; equality-constrained SOC test directions that preserve the constraints to first order. The bordered Hessian is a computational device for this test and is left to the formal Optimization course.

Further reading: Boyd, §19.2; MIT 6.7220, Lecture 7.

Output on a technology frontier

A firm chooses positive inputs (x, y) and production is $Q(x, y) = x^2 y$. The feasible frontier is

$$2x^2 + y^2 = 3.$$

The Lagrangian is

$$\mathcal{L} = x^2 y + \lambda(3 - 2x^2 - y^2),$$

with first-order conditions

$$2xy - 4\lambda x = 0, \qquad x^2 - 2\lambda y = 0, \qquad 2x^2 + y^2 = 3.$$

The interior FOCs give $x = y = 1$; boundary points with zero input give zero output on the nonnegative closure.

Shadow prices

With the sign convention $\mathcal{L} = f + \sum_j \lambda_j(b_j - g_j)$, and under regularity conditions ensuring a differentiable value function,

$$\lambda_j^* = \frac{\partial V}{\partial b_j}.$$

Thus the multiplier is the local marginal value of increasing the corresponding right-hand side b_j; when larger b_j means more resources, this is the value of relaxing the constraint.

- Consumer theory: λ is the marginal utility of income.
- Firm theory: λ is the marginal value of relaxing a resource constraint.

The Envelope theorem section derives this identity.

Restricted least squares

Constrained least-squares

Econometrics often imposes linear equality restrictions on coefficients:

$$y = X\beta + u, \qquad R\beta = r.$$

The constrained least-squares problem is

$$\min_{\beta} \, (y - X\beta)'(y - X\beta) \quad \text{s.t.} \quad R\beta = r.$$

Examples include equal treatment effects, zero long-run restrictions, and adding-up restrictions.

Likelihood ratio test

Restricted likelihood

A likelihood-ratio test compares the best unrestricted fit with the best fit under restrictions:

$$LR = 2\big(\ell(\widehat{\theta}) - \ell(\widehat{\theta}_R)\big).$$

Unrestricted. Maximize $\ell(\theta)$ over all admissible θ.
Restricted. Maximize $\ell(\theta)$ subject to $R\theta = r$.
The statistic measures how much optimized value is lost when the equality restrictions are imposed.

Section 4

Inequality constraints (KKT)

Binding or slack

Problem. $\max_{x \in \mathbb{R}^n} f(x)$ subject to $g_j(x) \leq b_j$.

At an optimum each constraint is either

Binding $g_j(x) = b_j$, or

Slack $g_j(x) < b_j$.

Resource constraint $x + y \leq 10$: plan $(4, 6)$ binds; plan $(3, 5)$ leaves two units unused and is slack.

KKT conditions

For $\max_x f(x)$ subject to $g_j(x) \leq b_j$, use $\mathcal{L}(x, \lambda) = f(x) + \sum_j \lambda_j (b_j - g_j(x))$. At a regular local optimum there are multipliers $\lambda^* \geq 0$ satisfying

Four conditions.

Stationarity	$\nabla_x \mathcal{L}(x^*, \lambda^*) = 0,$
Feasibility	$g_j(x^*) \leq b_j,$
Dual feasibility	$\lambda_j^* \geq 0,$
Complementary slackness	$\lambda_j^* (b_j - g_j(x^*)) = 0.$

Video: aa4cc, *Inequality-constrained optimization: KKT conditions*.

Complementary slackness

The arrows hold in only one direction, from inequality to equality: $g_j(x^*) < b_j \Rightarrow \lambda_j^* = 0$ and $\lambda_j^* > 0 \Rightarrow g_j(x^*) = b_j$. Do not reverse them: a binding constraint may still have zero multiplier.

Where this sits in duality theory

Suppose strong duality holds and both optima are attained. The primal and dual solutions are then a saddle point of the Lagrangian: the primal variables maximize it, the multipliers minimize it. The primal and dual problems, the dual function and strong duality are developed in the formal Optimization course.

Further reading: Boyd, Ch. 18; Boyd–Vandenberghe, §5.4.

Videos: Arizona Math Camp 40(A), conceptual and geometric insight; 40(B), conditions and theorem.

Nonnegativity: two equivalent KKT forms

Suppose the problem also requires $x_i \geq 0$. Write this as $-x_i \leq 0$ and let $\widetilde{\mathcal{L}}(x, \lambda) = f(x) + \sum_j \lambda_j \big(b_j - g_j(x)\big)$, omitting only the nonnegativity constraint.

With the multiplier μ_i.

$$\mathcal{L} = \widetilde{\mathcal{L}} + \mu_i x_i,$$

$$\frac{\partial \widetilde{\mathcal{L}}}{\partial x_i} + \mu_i = 0,$$

$$x_i \geq 0, \quad \mu_i \geq 0, \quad \mu_i x_i = 0.$$

Stationarity is an equality.

After eliminating μ_i. Since $\mu_i = -\partial\widetilde{\mathcal{L}}/\partial x_i$,

$$x_i \geq 0, \quad \frac{\partial \widetilde{\mathcal{L}}}{\partial x_i} \leq 0,$$

$$x_i \frac{\partial \widetilde{\mathcal{L}}}{\partial x_i} = 0.$$

If $x_i > 0$, the partial derivative is zero. If $x_i = 0$, it may be negative.

Further reading: Columbia Math Camp, *Constrained Optimization*, Sec. 6.

Minimization: the same system in the other sign convention

Standard minimization form

$$\min_x \quad f(x)$$
$$\text{s.t.} \quad g_i(x) \le 0, \qquad h_r(x) = 0.$$

$$\mathcal{L} = f(x) + \sum_i \lambda_i g_i(x) + \sum_r \nu_r h_r(x), \qquad \lambda_i \ge 0.$$

KKT system in the $g_i \le 0$ form.

$$\nabla f(x^*) + \sum_i \lambda_i^* \nabla g_i(x^*) + \sum_r \nu_r^* \nabla h_r(x^*) = 0,$$

$$g_i(x^*) \le 0, \quad h_r(x^*) = 0, \quad \lambda_i^* \ge 0, \quad \nu_r^* \in \mathbb{R}, \quad \lambda_i^* g_i(x^*) = 0.$$

Minimizing f is maximizing $-f$ on the same feasible set: this is the maximization system of the previous pages with the objective gradient and the multipliers relabelled, and $x_i \ge 0$ is handled by the same two equivalent forms.

Further reading: Boyd–Vandenberghe; MIT 6.7220.

Minimization: a solved KKT example

Problem and Lagrangian

$$\min_{x,y} \quad (x-1)^2 + (y-1)^2$$
$$\text{s.t.} \quad x+y-3 \geq 0, \qquad x, y \geq 0.$$

Let $c(x,y) = x + y - 3$. Then

$$\mathcal{L} = (x-1)^2 + (y-1)^2 - \lambda c(x,y) - \mu_x x - \mu_y y.$$

If $c(x,y) > 0$, then $\lambda = 0$. The nonnegativity KKT system then gives $(x,y) = (1,1)$, which is infeasible. Thus $c(x,y) = 0$.

Further reading: Tibshirani, CMU 10-725, *Karush-Kuhn-Tucker conditions*.

Minimization: KKT calculation

At the positive solution, $\mu_x = \mu_y = 0$. Stationarity and the binding constraint give

$$2(x-1)-\lambda = 0, \qquad 2(y-1)-\lambda = 0, \qquad x+y=3.$$

The first two equations and $x+y=3$ give

$$x = y = 1 + \frac{\lambda}{2}, \qquad 3 = x+y = 2+\lambda \quad \Rightarrow \quad \lambda = 1.$$

Substitution gives

$$\boxed{(x^*, y^*) = \left(\tfrac{3}{2}, \tfrac{3}{2}\right), \quad \lambda^* = 1, \quad \mu_x^* = \mu_y^* = 0.}$$

Primal and dual feasibility and complementary slackness hold. The objective is convex and the constraints define a convex feasible set, so this KKT point is the global minimizer.

Consumer choice: explicit multipliers

Consumer problem The consumer has perfect-substitutes utility:

$$\max_{x,y\geq 0} \quad \alpha x + \beta y$$
$$\text{subject to} \quad p_x x + p_y y \leq m,$$

where $\alpha, \beta, p_x, p_y, m > 0$.

Slack-form Lagrangian

$$\mathcal{L} = \alpha x + \beta y + \lambda(m - p_x x - p_y y) + \mu_x x + \mu_y y.$$

Here λ is the marginal utility of income; μ_x and μ_y enforce nonnegativity.

KKT conditions

Stationarity

$$\alpha - \lambda p_x + \mu_x = 0, \qquad \beta - \lambda p_y + \mu_y = 0.$$

Primal and dual feasibility

$$x, y \geq 0, \quad p_x x + p_y y \leq m, \qquad \lambda, \mu_x, \mu_y \geq 0.$$

Complementary slackness

$$\lambda(m - p_x x - p_y y) = 0, \qquad \mu_x x = \mu_y y = 0.$$

Consumer choice: reduced form

Eliminate each nonnegativity multiplier from stationarity and complementary slackness.

Good x

$$\alpha - \lambda p_x + \mu_x = 0,$$
$$\mu_x = \lambda p_x - \alpha \geq 0,$$
$$x(\alpha - \lambda p_x) = 0.$$

Good y

$$\beta - \lambda p_y + \mu_y = 0,$$
$$\mu_y = \lambda p_y - \beta \geq 0,$$
$$y(\beta - \lambda p_y) = 0.$$

Reduced KKT form.

$$\begin{aligned}
x &\geq 0, & \lambda p_x - \alpha &\geq 0, & x(\lambda p_x - \alpha) &= 0,\\
y &\geq 0, & \lambda p_y - \beta &\geq 0, & y(\lambda p_y - \beta) &= 0,\\
m - p_x x - p_y y &\geq 0, & \lambda &\geq 0, & \lambda(m - p_x x - p_y y) &= 0.
\end{aligned}$$

Consumer choice: corner solutions

Since $\alpha, \beta > 0$, utility is increasing in both goods, so the budget binds: $p_x x + p_y y = m$.

A consumed good pins down the multiplier: $x^* > 0 \Rightarrow \lambda^* = \alpha/p_x$ and $y^* > 0 \Rightarrow \lambda^* = \beta/p_y$.

At an x-corner, the unused good yields no more utility per dollar:

$$x^* > 0,\ y^* = 0 \quad \Rightarrow \quad \beta - \lambda^* p_y \leq 0 \quad \Leftrightarrow \quad \frac{\beta}{p_y} \leq \frac{\alpha}{p_x}.$$

Video: Economics in Many Lessons, *Utility maximization: a corner solution*.

Consumer choice: compare the two goods

Compare marginal utility per dollar

$$
\begin{aligned}
\frac{\alpha}{p_x} > \frac{\beta}{p_y} \quad &\Rightarrow \quad (x^*, y^*) = \left(\frac{m}{p_x}, 0\right), \\
\frac{\alpha}{p_x} = \frac{\beta}{p_y} \quad &\Rightarrow \quad \{(x, y) \in \mathbb{R}^2_+ : p_x x + p_y y = m\}, \\
\frac{\alpha}{p_x} < \frac{\beta}{p_y} \quad &\Rightarrow \quad (x^*, y^*) = \left(0, \frac{m}{p_y}\right).
\end{aligned}
$$

A KKT problem

Choose the feasible point closest to the target $(2,3)$:

$$\max_{x,y\geq 0}\ 10-[(x-2)^2+(y-3)^2]$$

subject to $x+y\leq 3$ and $x+2y\leq 6$.

- The unconstrained peak $(2,3)$ is infeasible.
- Optimum: $(x^*,y^*)=(1,2)$ and $(\lambda_1^*,\lambda_2^*)=(2,0)$.
- Constraint 2 and both nonnegativity constraints are slack at the optimum.

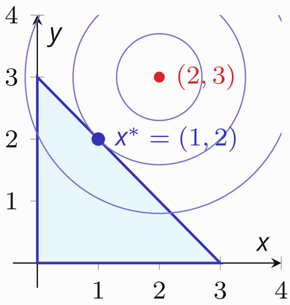

Figure 4 · Objective level circles and the feasible set.

The highest attainable level circle meets the feasible set at $x^*=(1,2)$.

Solving the KKT conditions by cases

When $x, y > 0$, the nonnegativity multipliers are zero and stationarity is $-2(x-2) - \lambda_1 - \lambda_2 = 0$ and $-2(y-3) - \lambda_1 - 2\lambda_2 = 0$. If a case reaches $x = 0$ or $y = 0$, the corresponding nonnegativity multiplier must be restored.

No resource constraint binds

If $\lambda_1 = \lambda_2 = 0$, interior stationarity gives $(x, y) = (2, 3)$, which violates $x + y \leq 3$.

Both resource constraints bind

If $\lambda_1, \lambda_2 > 0$, both constraints bind, so $(x, y) = (0, 3)$. Since $y > 0$, its nonnegativity multiplier is zero; y-stationarity requires $-\lambda_1 - 2\lambda_2 = 0$, impossible for positive multipliers.

One resource constraint binds

First constraint

If $\lambda_1 > 0$ and $\lambda_2 = 0$, stationarity gives

$$x = 2 - \frac{\lambda_1}{2}, \qquad y = 3 - \frac{\lambda_1}{2}.$$

Since $x + y = 3$,

$$5 - \lambda_1 = 3 \quad \Rightarrow \quad (x, y, \lambda_1) = (1, 2, 2),$$

and $x + 2y = 5 < 6$.

Second constraint

If $\lambda_1 = 0$ and $\lambda_2 > 0$, stationarity gives

$$x = 2 - \frac{\lambda_2}{2}, \qquad y = 3 - \lambda_2.$$

Since $x + 2y = 6$,

$$8 - \frac{5}{2}\lambda_2 = 6 \quad \Rightarrow \quad (x, y, \lambda_2) = \left(\frac{8}{5}, \frac{11}{5}, \frac{4}{5}\right),$$

but $x + y = 19/5 > 3$.

Surviving KKT point: $(x^*, y^*, \lambda_1^*, \lambda_2^*) = (1, 2, 2, 0)$; here $x, y > 0$, so the nonnegativity multipliers are indeed zero.

When KKT conditions are necessary

Under differentiability and a **constraint qualification**, a local optimum satisfies KKT.

Further reading: Boyd–Vandenberghe, Ch. 5; MIT 6.7220, Lec. 7.

When KKT conditions are sufficient

If (x^*, λ) satisfies KKT, f is concave, each g_j is convex, and the sign convention is $\mathcal{L} = f + \sum_j \lambda_j (b_j - g_j)$, then x^* solves the problem globally.

(One of various versions of the sufficient condition)

Convex g_j have convex sublevel sets $\{x : g_j(x) \leq b_j\}$, so their intersection is convex; concavity of f turns the first-order inequalities into a global conclusion.

KKT sufficiency beyond concavity

Quasiconcave programming

For the convention $g_j(x) \leq b_j$, suppose f and the g_j are differentiable and each g_j is quasiconvex. If (x^*, λ^*) satisfies KKT, then x^* is a global maximizer when either

(a) f is concave; or

(b) f is quasiconcave and $\nabla f(x^*) \neq 0$.

The requirement $\nabla f(x^*) \neq 0$ in (b) is imposed at the KKT point. It excludes stationary points at which quasiconcavity gives no first-order ranking of feasible alternatives.

Further reading: Walker, *Nonlinear Programming and the Kuhn–Tucker Conditions*.

Pseudoconcavity and first-order sufficiency

Pseudoconcavity

On a convex domain S, a differentiable $f : S \to \mathbb{R}$ is pseudoconcave when, for every $x, y \in S$,

$$f(y) > f(x) \quad \Rightarrow \quad \nabla f(x)^\top (y - x) > 0.$$

This implication directly converts the KKT gradient inequality for a feasible y into $f(y) \leq f(x^*)$. Hence a pseudoconcave objective and quasiconvex constraint functions also make KKT sufficient.

Pseudoconcavity: assumptions used by the theorem

Gradient conditions

Walker's KKT theorem needs $\nabla f(x^*) \neq 0$ only at the KKT point. His separate characterization of pseudoconcavity from quasiconcavity assumes $\nabla f(x) \neq 0$ throughout the domain.

Further reading: Walker, *Differentiable Quasiconcave Functions*.

Constraint qualifications

Constraint qualifications

Constraint qualifications rule out irregular local descriptions of the feasible boundary. For this brush-up, assume that a suitable condition holds.
In Optimization you will study constraint qualifications.
Further reading: Boyd, §19.4; MIT 6.7220, Lec. 7;
Border, *Summary Notes on Maximization*, §6, pp. 11–12.

Video: Stanford EE364A, *Lecture 8: Lagrangian duality and Slater's condition*.

Complementary slackness and shadow prices

Let

$$V(b) = \max_x f(x) \qquad \text{s.t.} \qquad g_j(x) \le b_j,$$

and use $\mathcal{L} = f + \sum_j \lambda_j (b_j - g_j)$. Under regularity and differentiability of the value function,

$$\frac{\partial V}{\partial b_j} = \lambda_j^*.$$

Which constraints have a positive shadow price

Complementary slackness ties each multiplier to the value of relaxing its constraint:

- If $g_j(x^*) < b_j$, then $\lambda_j^* = 0$, so $\partial V/\partial b_j = 0$: relaxing an unused constraint has zero first-order value.
- If $\lambda_j^* > 0$, then $g_j(x^*) = b_j$: the constraint binds and relaxing its right-hand side has positive marginal value.
- **Binding alone does not imply positive marginal value; a binding constraint may have $\boldsymbol{\lambda_j^* = 0}$.**

Shadow prices in the two-resource example

In the two-resource example, $\lambda_1^* = 2$ and $\lambda_2^* = 0$: a marginal relaxation of the first resource constraint is worth 2, while the slack second constraint has zero shadow price.

The **parameter-envelope formula** for KKT problems is derived in the Envelope theorem section, after comparative statics separates movements in the optimizer from movements in optimized value.

Section 5

Comparative statics

Parameterized problems: solution and value

For a parameter θ, let the feasible set be $S(\theta)$ and the objective be $f(x,\theta)$. Two distinct objects are

$$C(\theta) = \arg\max_{x \in S(\theta)} f(x,\theta), \qquad V(\theta) = \max_{x \in S(\theta)} f(x,\theta).$$

$C(\theta)$ is the **solution correspondence**: it may contain several maximizers.
$V(\theta)$ is the **value function**: when the maximum is attained, it equals the optimized objective value.

Consumer demand and indirect utility

A consumer with prices $p \gg 0$ and wealth w has

$$B(p,w) = \{x \in \mathbb{R}^n_+ : p \cdot x \leq w\}, \qquad C(p,w) = \arg\max_{x \in B(p,w)} u(x), \qquad V(p,w) = \max_{x \in B(p,w)} u(x).$$

If u is strictly quasiconcave on the convex budget set, $C(p,w)$ is a singleton whenever it is nonempty and can be written as an ordinary demand function. The optimizer set $C(p,w)$ and optimized value $V(p,w)$ are different objects.

Continuity properties of C and V belong to the later optimization course.

Two sensitivity questions

For a parameterized problem, the solution object $C(\theta)$ and value $V(\theta)$ answer different questions.

Comparative statics.
How does the optimal choice change with the parameter? If the solution is unique and differentiable,

$$\theta \mapsto x^*(\theta), \qquad \frac{dx^*}{d\theta}.$$

Using the implicit function theorem.

Envelope theorem.
How does the optimized value change with the parameter?

$$V(\theta) = f(x^*(\theta), \theta), \qquad \frac{dV}{d\theta}.$$

At a regular optimum, the envelope theorem often avoids computing $dx^*/d\theta$.

Same parameterized problem, two different derivatives: one tracks the optimizer; the other tracks the optimized value.

Video: Matt Clancy, *Comparative statics*.

Local derivative of an optimizer

Differentiate the first-order condition

Suppose an interior optimizer satisfies

$$F(x^*(\theta), \theta) = 0,$$

and $D_x F$ is invertible at the point of interest. The implicit function theorem gives

$$\frac{dx^*}{d\theta} = -[D_x F(x^*, \theta)]^{-1} D_\theta F(x^*, \theta).$$

For $F = \nabla_x f$, the matrix $D_x F$ is the Hessian with respect to x.

Further reading: Boyd, Ch. 15; §19.3; Border, *Implicit Function Theorem*.

Firm supply from an FOC

Interior output choice

A price-taking firm chooses an interior output $q^*(p)$:

$$\max_{q>0} \; pq - c(q), \qquad c''(q) > 0.$$

The first-order condition is

$$p - c'(q^*(p)) = 0.$$

Price and local supply

Differentiating the firm's first-order condition

Differentiate this identity with respect to p:

$$1 - c''(q^*(p))\,\frac{dq^*}{dp} = 0.$$

Hence

$$c''(q^*(p))\,\frac{dq^*}{dp} = 1, \qquad \boxed{\frac{dq^*}{dp} = \frac{1}{c''(q^*(p))} > 0.}$$

This is the implicit function theorem from Differentiation applied to an optimization FOC: with strictly convex cost, a higher output price raises the firm's optimal output locally.

Increasing differences

Monotone comparative statics

For problems with corners or discrete choices, increasing differences ranks the gain from a higher x across values of θ:

$$x' > x, \quad \theta' > \theta \quad \Rightarrow \quad f(x', \theta') - f(x, \theta') \geq f(x', \theta) - f(x, \theta).$$

Higher θ raises the gain from higher x.
The broader toolkit includes supermodularity, single crossing, lattices, and Topkis-type results.

Further reading: Echenique, Monotone Comparative Statics.

Section 6

Envelope theorem

Why the indirect effect disappears

Let an interior optimizer $x^*(\alpha)$ define

$$V(\alpha) = \max_x f(x, \alpha) = f(x^*(\alpha), \alpha).$$

The chain rule gives

$$V'(\alpha) = f_x(x^*(\alpha), \alpha)\frac{dx^*}{d\alpha} + f_\alpha(x^*(\alpha), \alpha).$$

At an interior optimum, the FOC is

$$f_x(x^*(\alpha), \alpha) = 0.$$

Therefore

$$\begin{aligned} V'(\alpha) &= 0 \cdot \frac{dx^*}{d\alpha} + f_\alpha(x^*(\alpha), \alpha) \\ &= \boxed{f_\alpha(x^*(\alpha), \alpha)}. \end{aligned}$$

Two derivatives from one parameterized problem

Comparative statics asks for $dx^* / d\alpha$. The envelope theorem asks for $dV/d\alpha$ and shows why that derivative can often be computed without solving for $dx^* / d\alpha$: the indirect first-order term vanishes at an interior optimum.

Further reading: Boyd, §19.1. Video: Arizona Math Camp, *The Envelope Theorem and Shadow Values*.

Equality-constrained derivation

Consider

$$V(\alpha) = \max_x f(x, \alpha) \quad \text{s.t.} \quad g(x, \alpha) = 0, \qquad \mathcal{L} = f + \lambda(0 - g).$$

Along a differentiable regular solution path,

$$V' = f_x x^{*\prime} + f_\alpha.$$

The Lagrange FOC and the derivative of the binding constraint give

$$f_x = \lambda^* g_x, \qquad g_x x^{*\prime} + g_\alpha = 0 \quad \Rightarrow \quad g_x x^{*\prime} = -g_\alpha.$$

Substituting,

$$\begin{aligned} V' &= \lambda^* g_x x^{*\prime} + f_\alpha \\ &= f_\alpha - \lambda^* g_\alpha \\ &= \boxed{\mathcal{L}_\alpha(x^*, \lambda^*, \alpha)}. \end{aligned}$$

The direct parameter effect remains, together with the parameter's effect on the binding constraint valued at the shadow price.

Inequality-constrained envelope

Consider a regular KKT problem in which the objective, constraints, and right-hand sides may depend on a scalar parameter α:

$$V(\alpha) = \max_x f(x, \alpha) \qquad \text{s.t.} \qquad g_j(x, \alpha) \leq b_j(\alpha).$$

With

$$\mathcal{L} = f + \sum_j \lambda_j (b_j - g_j),$$

along a differentiable solution with a locally unchanged active set,

$$\boxed{V'(\alpha) = \mathcal{L}_\alpha(x^*, \lambda^*, \alpha)},$$

The parameter derivative

$$\mathcal{L}_\alpha = f_\alpha(x^*, \alpha) + \sum_j \lambda_j^* \left(b_j'(\alpha) - g_{j,\alpha}(x^*, \alpha)\right).$$

- If only the objective depends on α, then $V'(\alpha) = f_\alpha(x^*, \alpha)$.
- If the parameter is a right-hand side b_j, then $\partial V/\partial b_j = \lambda_j^*$, recovering the shadow-price formula from KKT.

KKT, parameter changes & envelope

For $\alpha \geq 0$, consider

$$\sup_{x,y>0} \ \alpha \ln x + \ln y$$

subject to

$$2x + y \leq 6.$$

(a) For $\alpha > 0$, write the KKT conditions and solve for $x^*(\alpha)$ and $y^*(\alpha)$.

(b) At $\alpha = 0$, determine whether a maximizer exists and find the supremum.

(c) Let $V(\alpha)$ be the supremum. Use the envelope theorem to determine where V decreases and increases.

(d) Add $x \leq y$. Determine whether the solution changes at $\alpha = 1$ and $\alpha = 3$; find the revised solution when it changes.

KKT, parameter changes and envelope

KKT and envelope calculation. With multiplier $\lambda \geq 0$,

$$\mathcal{L} = \alpha \ln x + \ln y + \lambda(6 - 2x - y).$$

The KKT conditions are

$$\frac{\alpha}{x} = 2\lambda, \qquad \frac{1}{y} = \lambda, \qquad \lambda \geq 0, \quad 2x + y \leq 6, \quad \lambda(6 - 2x - y) = 0.$$

The objective is increasing in both variables, so the resource constraint binds: $2x + y = 6$. The first-order conditions imply

$$x = \frac{\alpha}{2\lambda}, \qquad y = \frac{1}{\lambda}, \qquad 2x + y = \frac{\alpha + 1}{\lambda} = 6.$$

Hence

$$x^*(\alpha) = \frac{3\alpha}{\alpha + 1}, \qquad y^*(\alpha) = \frac{6}{\alpha + 1}, \qquad \lambda^*(\alpha) = \frac{\alpha + 1}{6}.$$

KKT, parameter changes and envelope

At $\alpha = 0$, the objective is $\ln y$. Feasibility with $x > 0$ implies $y < 6$, so no maximizer exists and

$$V(0) = \sup_{x,y>0,\ 2x+y\leq 6} \ln y = \ln 6.$$

For $\alpha > 0$,

$$V(\alpha) = \ln 6 + \alpha \ln 3 + \alpha \ln \alpha - (\alpha+1)\ln(\alpha+1).$$

The envelope theorem gives

$$V'(\alpha) = \ln x^*(\alpha) = \ln\left(\frac{3\alpha}{\alpha+1}\right).$$

Since $V(\alpha) \to \ln 6$ as $\alpha \downarrow 0$, V is continuous at zero. It decreases on $[0, 1/2]$ and increases on $[1/2, \infty)$.

KKT, parameter changes and envelope

For $\alpha > 0$,

$$\frac{x^*(\alpha)}{y^*(\alpha)} = \frac{\alpha}{2}.$$

At $\alpha = 1$, $(x^*, y^*) = (3/2, 3)$, so $x \leq y$ is satisfied. At $\alpha = 3$, $(x^*, y^*) = (9/4, 3/2)$, so the added constraint changes the solution. The binding resource constraint gives $y = 6 - 2x$, and $x \leq y$ becomes $x \leq 2$. Strict concavity then places the revised solution at

$$(x^*, y^*) = (2, 2).$$

Roy's identity

Roy's identity from the value function

Let

$$v(p,w) = \max_{x \geq 0}\{u(x) : p \cdot x \leq w\}.$$

Under standard differentiability and interior regularity conditions, with $v_w(p,w) > 0$, Marshallian demand satisfies

$$x_i(p,w) = -\frac{v_{p_i}(p,w)}{v_w(p,w)}.$$

This is the envelope theorem applied to the consumer's value function.

Video: Economics in Many Lessons, *Roy's identity*.

Hotelling and Shephard

Envelope identities

Define the profit and cost functions by

$$\pi(p) = \max_{y \in Y} p \cdot y, \qquad c(w,q) = \min_{x \geq 0}\{w \cdot x : F(x) \geq q\}.$$

Under standard differentiability and interior regularity conditions:

Hotelling	supply from profit	$\dfrac{\partial \pi(p)}{\partial p_i} = y_i(p),$
Shephard	conditional factor demand from cost	$\dfrac{\partial c(w,q)}{\partial w_i} = x_i(w,q).$

Videos: Arizona Math Camp, *Hotelling's lemma*; *Shephard's lemma*.

Convex duality behind the value functions

Expenditure, cost, and profit

indicator of a feasible set $\Rightarrow$ conjugate / support function

concave conjugates: $e(p, \bar{u})$, $c(w, q)$ convex conjugate: $\pi(p)$

Degree-one homogeneity in p or w.

Shephard and Hotelling identities; demand/supply monotonicity.

Hessian sign restrictions; recovery of closed convex sets.

Study strategy. When you cannot memorize all these properties in Micro I, derive them from the support function theorem.

Further reading: Boyd, Ch. 7, §§7.4.3–7.5.6 (pp. 29–39); Ch. 13, §13.3.5 (pp. 24–25).

Auction payment formula

Envelope characterization

In a single-object auction, type v wins with probability $q(v)$ and pays $p(v)$, so $U(v) = vq(v) - p(v)$.

Under the standard one-dimensional incentive-compatibility conditions, U is absolutely continuous and

$$U'(v) = q(v) \text{ a.e.}, \qquad U(v) = U(\underline{v}) + \int_{\underline{v}}^{v} q(t)\,dt.$$

Since $U(v) = vq(v) - p(v)$,

$$p(v) = vq(v) - U(\underline{v}) - \int_{\underline{v}}^{v} q(t)\,dt.$$

Once the allocation rule is fixed, payments are pinned down up to the lowest type's surplus.

Further reading: Roughgarden, auction notes.

Section 7

Worked example

Cobb-Douglas utility maximization

Let $\alpha, \beta, p_1, p_2, w > 0$. Maximize $x^\alpha y^\beta$ subject to $p_1 x + p_2 y \leq w$ and $x, y > 0$. Monotonicity makes the budget bind. By the log trick, maximize $\alpha \ln x + \beta \ln y$ instead, with

$$\mathcal{L} = \alpha \ln x + \beta \ln y + \lambda(w - p_1 x - p_2 y).$$

The first-order conditions are

$$\frac{\alpha}{x} = \lambda p_1, \qquad \frac{\beta}{y} = \lambda p_2, \qquad p_1 x + p_2 y = w.$$

Thus

$$p_1 x = \frac{\alpha}{\lambda}, \qquad p_2 y = \frac{\beta}{\lambda}.$$

Video: Economics in Many Lessons, *Utility Maximization with a Cobb-Douglas Utility Function*.

Cobb-Douglas utility maximization: solve the conditions

Adding the two expenditure equations and using the binding budget gives

$$w = p_1 x + p_2 y = \frac{\alpha + \beta}{\lambda} \quad \Rightarrow \quad \lambda^* = \frac{\alpha + \beta}{w}.$$

Substitution into the first-order conditions yields

$$x^* = \frac{\alpha}{\alpha + \beta}\frac{w}{p_1}, \qquad y^* = \frac{\beta}{\alpha + \beta}\frac{w}{p_2}.$$

The log objective $\alpha \ln x + \beta \ln y$ is strictly concave on $\mathbb{R}^2_{++}$, and the budget set is convex. Hence the regular first-order conditions identify the unique global maximizer.

Further reading: Border, *Preference and Demand Examples*, Sections 1–2.

Expenditure shares and the multiplier

Expenditure shares

Demands spend fixed income shares $\alpha/(\alpha+\beta)$ and $\beta/(\alpha+\beta)$ on the two goods.

Multiplier

$$\lambda^* = \frac{\alpha+\beta}{w} = \widetilde{V}_w.$$

It is the envelope derivative of the indirect log-value.

Section 8

Further reading

Optimization references

Recommended references

Osborne, *Mathematical Methods for Economic Theory*, optimization sections [link].

Echenique, *Constrained Maximization* [link].

Border, *Miscellaneous Notes on Optimization Theory and Related Topics* [link].

Comparative statics and extensions

Selected references

Echenique, *Monotone Comparative Statics* [link];

Border, *Envelope Theorem* [link].

Border, *Single Crossing Properties* [link] and *Lattices and Supermodularity* [link].

Sarver, *Microeconomic Theory Lecture Notes*, Chapters 1–2 [link].

Original sources: Milgrom and Shannon (1994) [link]; Topkis (1998) [link].